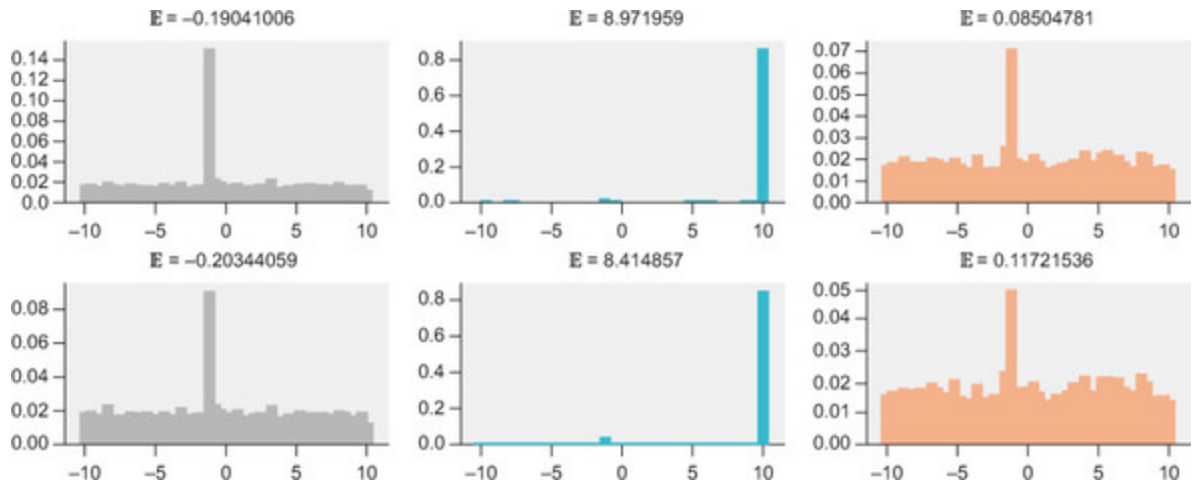
Figure 7.22 shows a few of the learned distributions in the experience replay buffer, to give you a better view of the distributions. Each row is a sample from the replay buffer associated with a single state. Each figure in a row is the action-value distribution for the NO-OP, UP, and DOWN actions respectively. Above each figure is the expected value of that distribution. You can see that in all the samples, the UP action has the highest expected value, and it has two clear peaks: one at –1 and another at +10. The distributions for the other two actions have a lot more variance, because once the agent learns that going up is the best way to win, there are fewer and fewer experiences using the other two actions, so they remain relatively uniform. If we continued training for longer, they would eventually converge to a peak at –1 and possibly a smaller peak at –10, since with epsilon greedy we will still be taking a few random actions.

Figure 7.22. Each column has the action-value distributions for a particular action for a given state (row). The number above each plot is the expectation value for that distribution, which is the weighted average value for that distribution. The distributions look fairly similar by eye, but the expected values are distinct enough to result in significantly different action selections.



Distributional Q-learning is one of the biggest improvements to Q-learning in the past few years, and it's still being actively researched. If you compare Dist-DQN to ordinary DQN, you should find you get better overall performance with Dist-DQN. It is not well understood why Dist-DQN performs so much better, especially given that we are only choosing actions based on expected values, but a few reasons are likely. One is that training a neural network to predict multiple things at the same time has been shown to improve generalization and overall performance. In this chapter, our